What Downstream Scores Miss in Point-Cloud Self-Supervised Learning

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github.com/KohsukeIde/WhatScoresMiss

A downstream score says how well, not why

Downstream score: the benchmark result after pretraining (ScanNet, ScanObjectNN, ShapeNetPart).
A higher score can come from any of three stages:

1 · Pretraining
What signal is learned?

→

2 · Test input
What geometry is visible?

→

3 · Reported score
What does the score measure?

We test each stage with a controlled diagnostic.

Issue 1: did the model need the signal the pretext task targets?

1 · Pretraining

What signal is learned?

→

2 · Test input

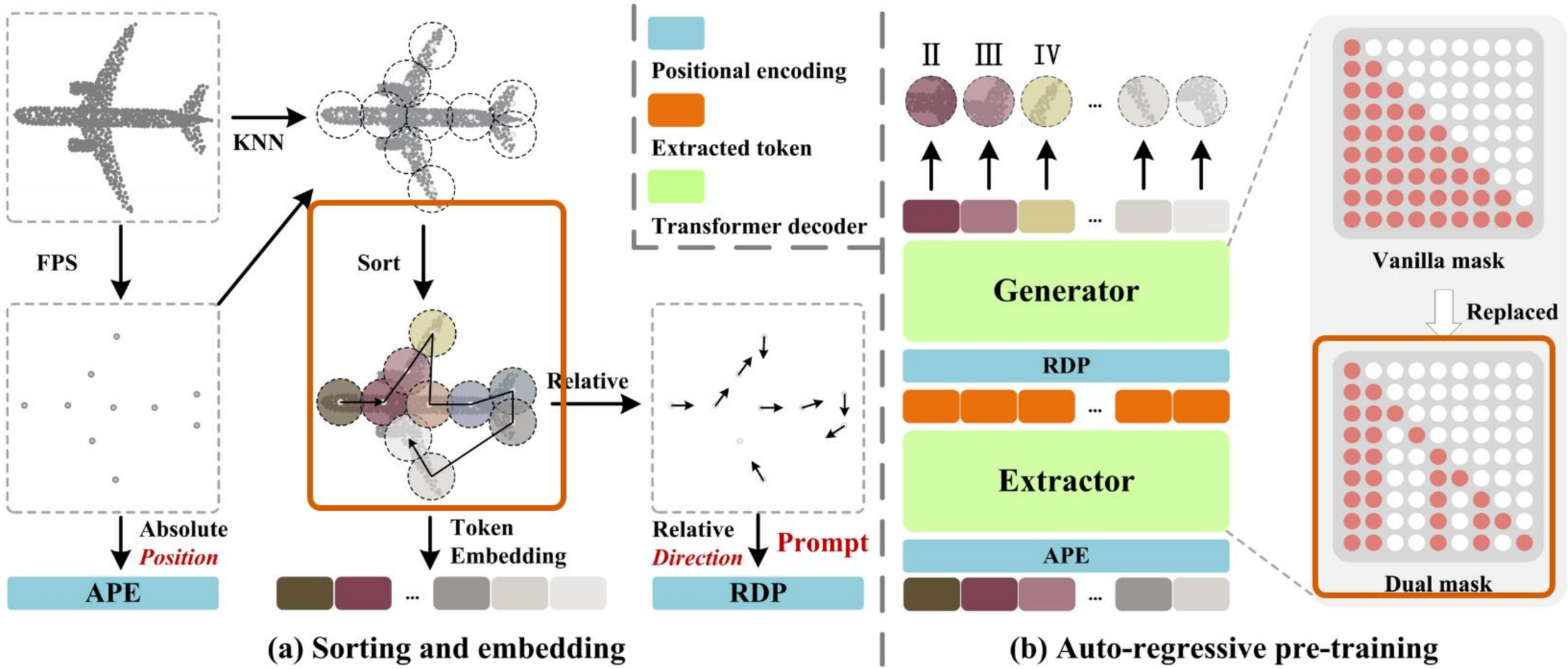
What geometry is visible?

→

3 · Reported score

What does the score measure?

PointGPT is built on a patch order and a dual mask



Are both needed? We trained variants without the dual mask, with a random order, or both.

Figure: PointGPT (Chen et al., NeurIPS 2023).

Without the mask or the order, transfer stays high

PointGPT variant	Pretext loss ↓ (%)	obj_bg	obj_only	PB_T50_RS	ShapeNetPart
Reference: dual mask, sorted order	–	90.36	90.53	86.68	85.66
No dual mask	90.66	89.21	89.38	84.19	85.60
Random order	88.72	90.02	87.95	**83.90 (–2.78)**	**85.44 (–0.22)**
No dual mask, random order	89.02	90.53	88.30	83.97	85.55

ScanObjectNN top-1 accuracy (%) on its three variants; ShapeNetPart instance mIoU (%). Pretext loss ↓: within-run decrease from the first to the final checkpoint.

Largest drop: 2.78 points (Random order, PB_T50_RS); 0.22 on ShapeNetPart.

Paper Table 2.

Issue 2: what does missing-point robustness show?

1 · Pretraining

What signal is learned?

→

2 · Test input

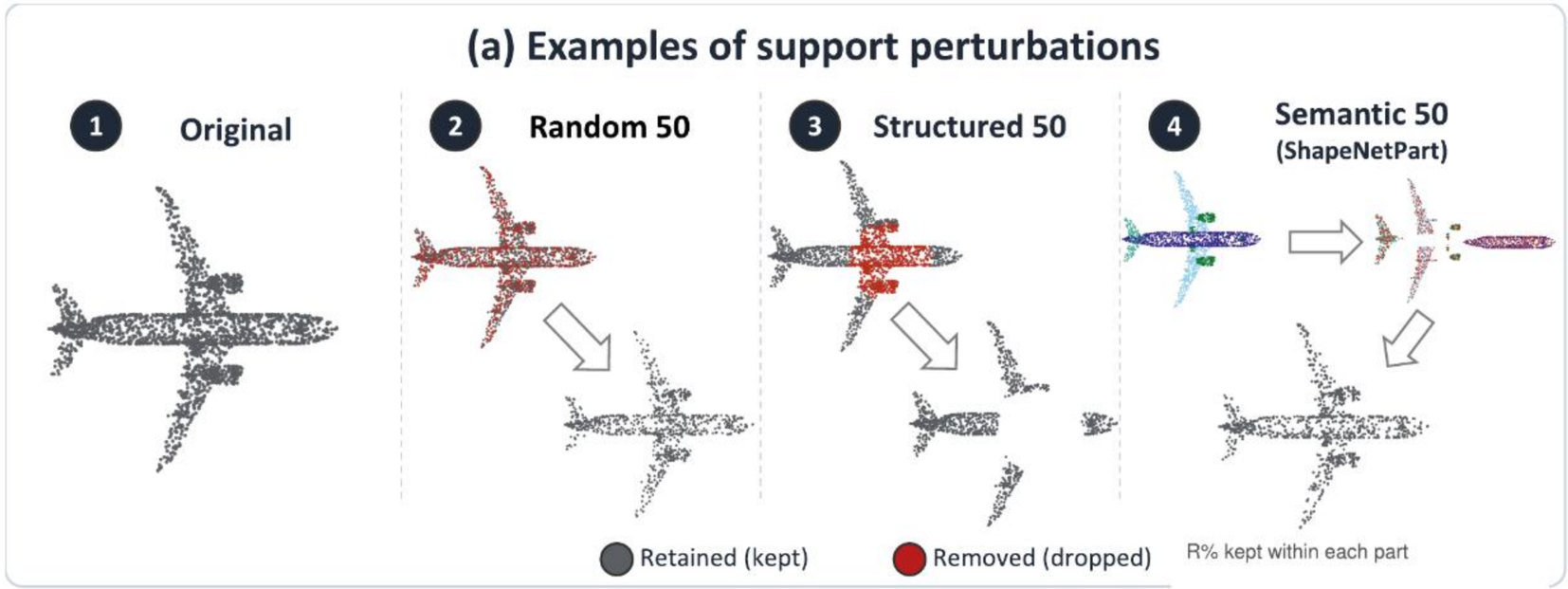
What geometry is visible?

→

3 · Reported score

What does the score measure?

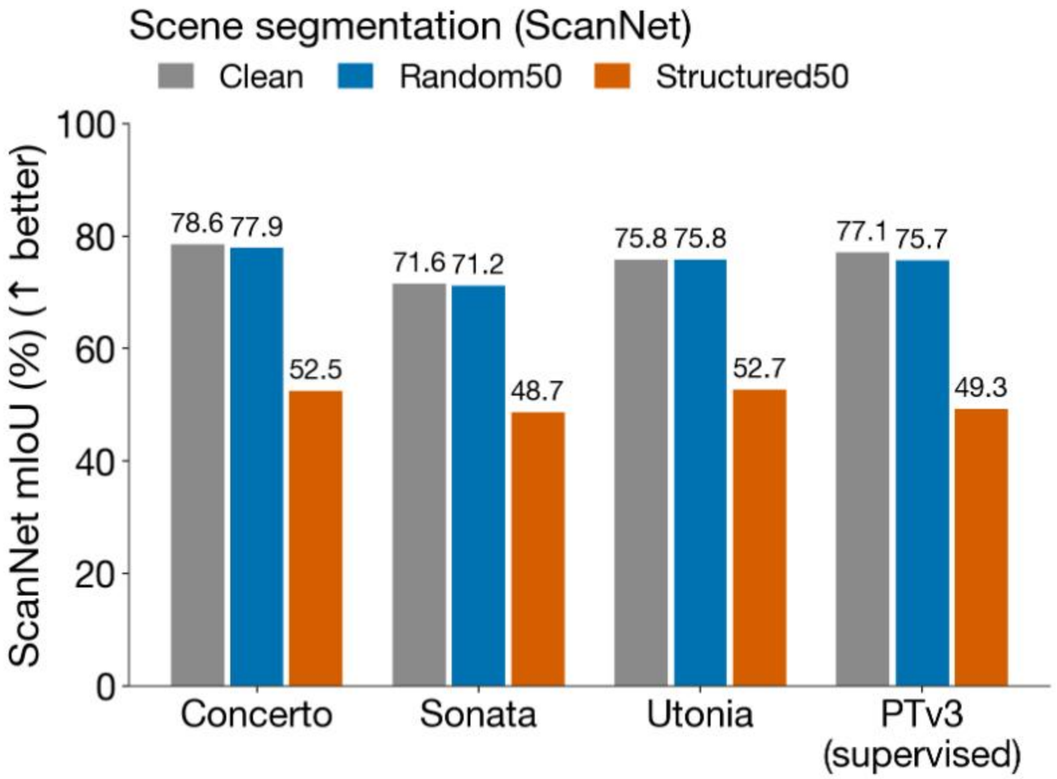
Random removal keeps the shape; structured removal does not

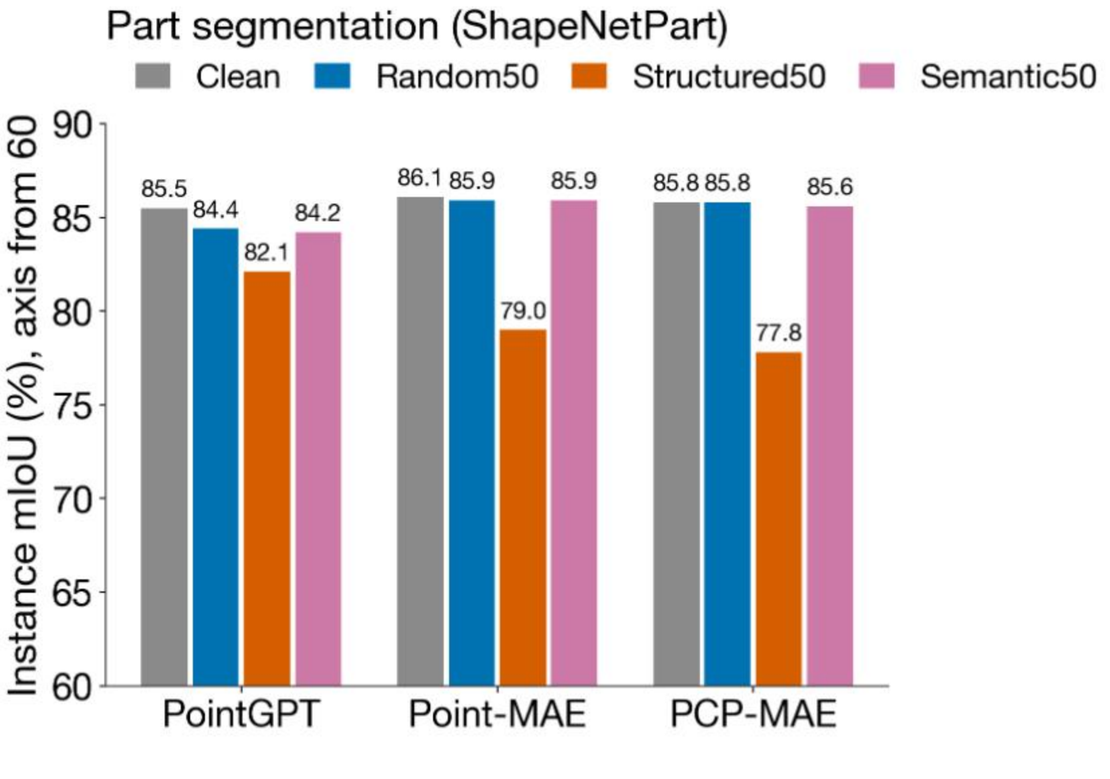


Same number of points, different visible geometry.

Paper Fig. 2. Structured50 keeps whole spatial cells (scenes) or the 50% of points farthest from a random anchor (objects). Semantic50 keeps 50% of each part.

With half the points kept, structured removal hurts far more

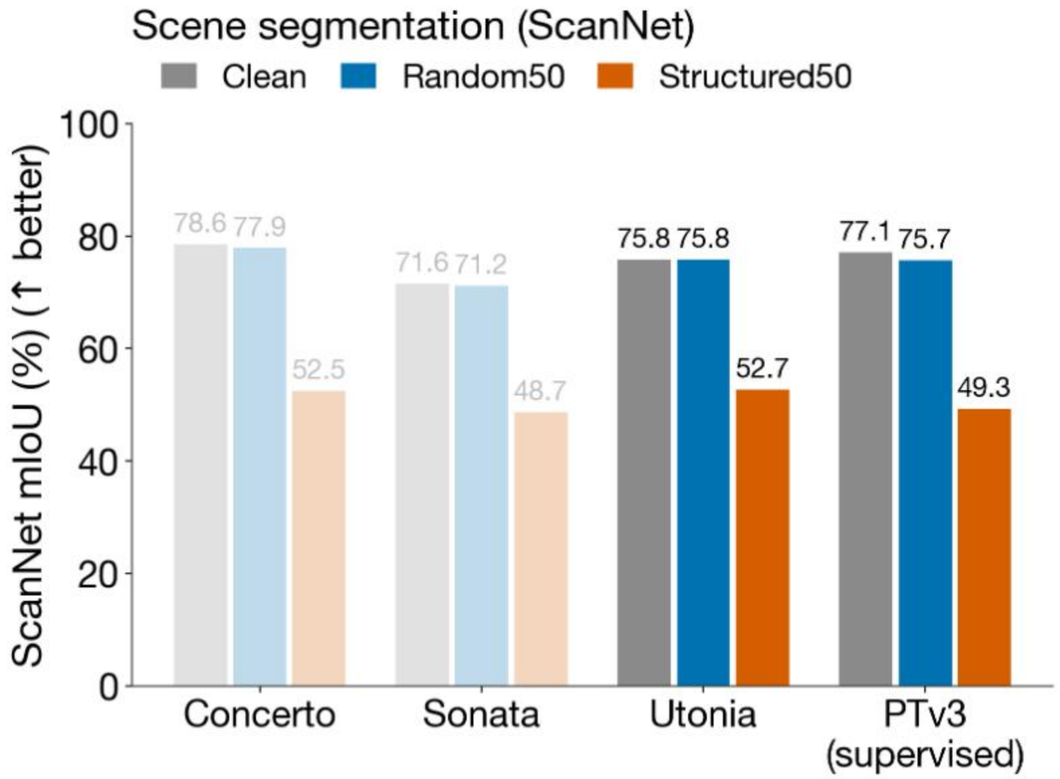


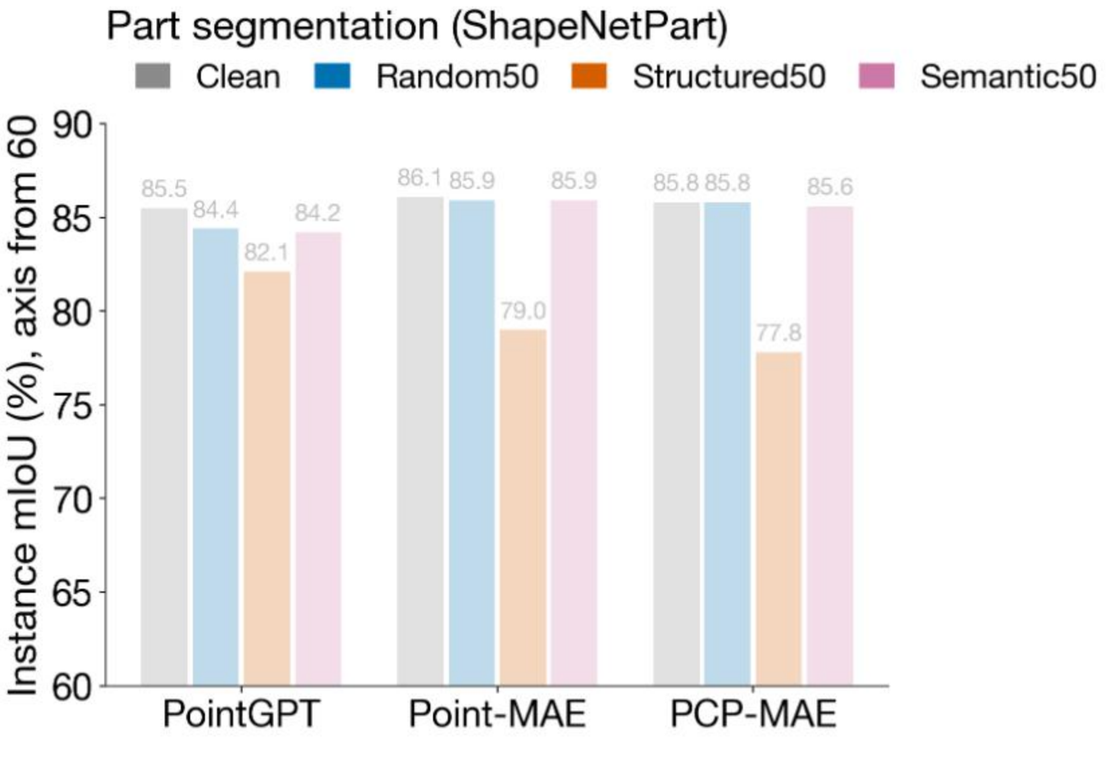


Random50 stays close to clean for every model.

Paper Fig. 4. Scene models: Concerto, Sonata, Utonia (SSL) and PTv3 (supervised). Scene scores propagate labels from the kept points to the full scene.

Under structured removal, the model ranking can flip





Clean: PTv3 77.1 > Utonia 75.8 · Structured50: PTv3 49.3 < Utonia 52.7

Paper Fig. 4. Scene models: Concerto, Sonata, Utonia (SSL) and PTv3 (supervised). Scene scores propagate labels from the kept points to the full scene.

Issue 3: what does a reported score leave unresolved?

1 · Pretraining

What signal is learned?

→

2 · Test input

What geometry is visible?

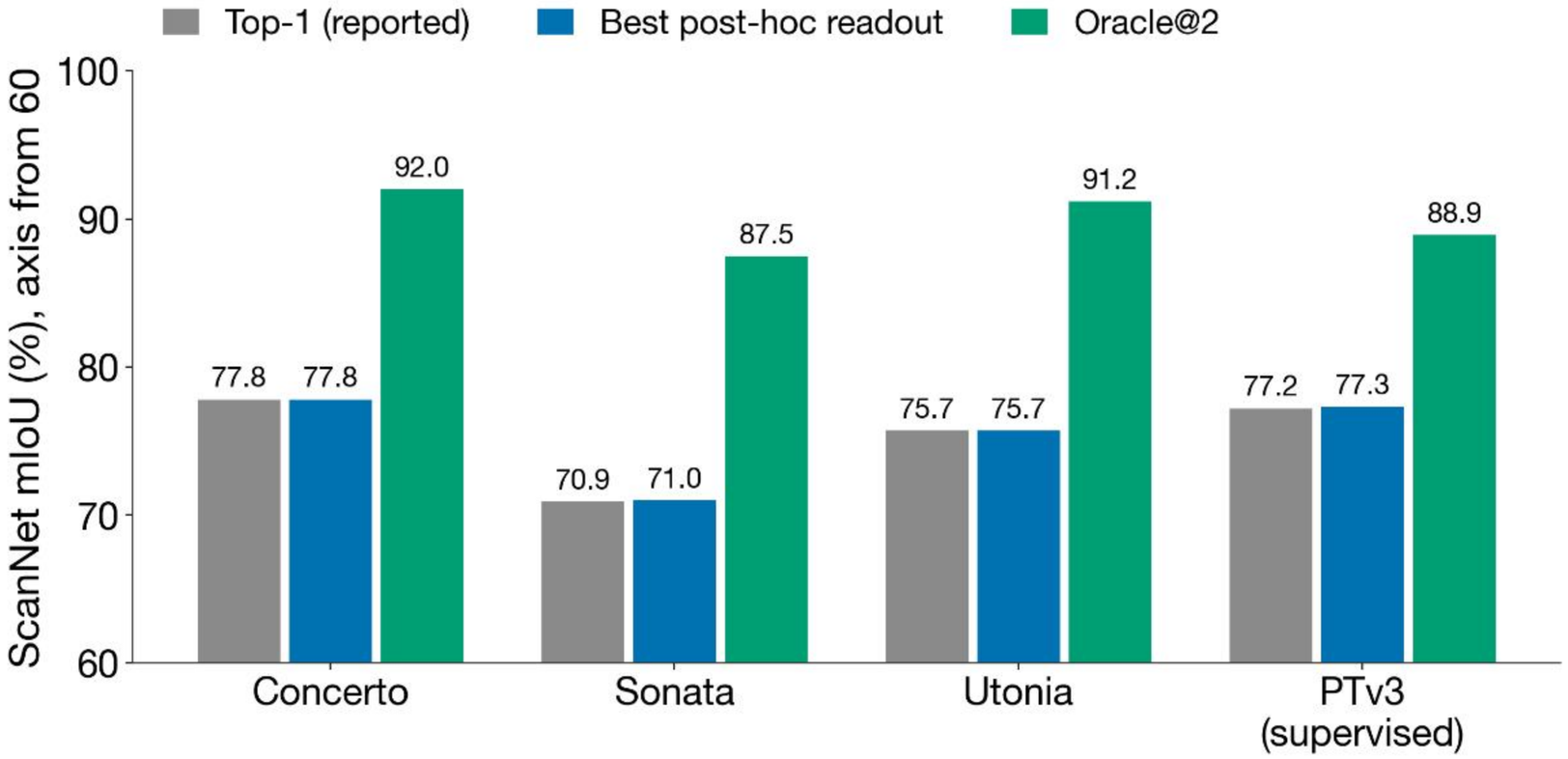
→

3 · Reported score

What does the score measure?

Top-1 mIoU hides how close the right class was

Oracle@2: a point counts as correct if the true class is among the top-2 predictions. A diagnostic of how close the right class was, not a prediction method.



Top-1 cannot tell a near miss from a far one.

Paper Table 5. Best post-hoc readout: the best of the readouts refit on the same frozen features.

Treat a downstream gain as a claim, and test its mechanism

✓ Downstream scores miss side channels in pretraining, which points the task needs, and how the score is formed.

→ What to report:

> **Pretext objective** State the intended signal and test its side channels.

> **Input support** Test beyond random removal, with the same number of points.

> **Reported score** Report Oracle@k and post-hoc readout checks.

> **Selection** State how checkpoints and splits were chosen.

✗ Scope: indoor scenes and ShapeNet objects. Contrastive SSL and outdoor LiDAR are future work.

Code
github.com/KohsukeIde/WhatScoresMiss

Project page
kohsukeide.github.io/projects/what-scores-miss